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Bihar Engineering University, Patna

B.Tech 3rd Semester Examination, 2023

Time: 03 Hours

Code: 100302 ***Subject: Analog Electronic Circuits*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Answer the following questions (any seven only): **[14]**

(a) Mention the advantages of Full Wave Rectifier.

(b) Draw the pn junction diode VI characteristics.

(c) What are the different configurations of BJT?

(d) What is stability factor?

(e) What is MOSFET? Classify the types of MOSFET

(f) What is meant by pinchoff voltage?

(g) Define an operational amplifier.

(h) Define Slew rate.

(i) Mention some of the non-linear applications of op-amp.

(j) List the features of instrumentation amplifier.

Q.2 Solve both questions : **[14]**

(a) What is a PN Junction? Explain the formation of depletion layer in a PN junction.

(b) Draw the circuit diagram of half wave rectifier and explain its operation with the help of waveforms.

Q.3 Solve both questions : **[14]**

(a) If the base current in a transistor is 20 μA when the emitter current is 6.4mA, what are the values of α and β? Also calculate the collector current.

(b) Why hybrid model is used for the analysis of BJT amplifier at low frequencies? Draw the hybrid model for CE transistor and derive the parameters.

Q.4 Solve both questions : **[14]**

(a) With the help of neat diagram, explain the operation and characteristics of n-channel enhancement type MOSFET.

(b) Draw and explain the small signal model of FET at low frequency.

Q.5 Solve both questions : **[14]**

(a) What are the factors that affect the frequency stability of an oscillator? How can be improved in oscillators.

(b) Compare different configurations of differential amplifier.

Q.6 Solve both questions : **[14]**

(a) Write the design steps of the first order low pass filter and draw its circuit.

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(b) With the help of neat and clear sketch, explain the working of a MOS transistor as a switch.

Q.7 Solve this question : **[14]**

Explain the basic building block diagram of an op-amp. Write short note on input bias current, output bias current, slew rate and output offset voltage of an op-amp.

Q.8 Solve both questions : **[14]**

(a) Explain the working of series clipping circuit to clip the input sinusoidal signal above reference level. Draw the waveforms and transfer characteristics.

(b) Compare BJT and FET on different parameters.

Q.9 Solve both questions : **[14]**

(a) Write Short notes on (i) Zero Crossing Detector (ii) Precision Rectifier

(b) Explain the function of Square-wave and triangular-wave generators.

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